Course Information Sheet

Thermodynamics (ET2113)

Document No. CRS-0075 · Academic Year 2024/2025 · Semester 1

Course Details

Field	Details
Course ID	18
Course Name	Thermodynamics
Course Code	ET2113
Credit Value	3.00
Department	Engineering Technology (Department ID: 2)

Course Description

This course develops the first and second laws of thermodynamics and their application to engineering technology. Students study the properties of pure substances, energy transfer by work and heat, ideal gas processes, entropy, and the analysis of power and refrigeration cycles. The behaviour of real systems is examined through illustrative examples drawn from engines, compressors, and thermal plant. Problem classes emphasize systematic application of the laws to open and closed systems.

Learning Outcomes

On successful completion of this course, students will be able to:

- State and apply the first and second laws of thermodynamics to closed and open systems.
- Determine the properties of pure substances and model ideal gas processes.
- Analyse basic power and refrigeration cycles and interpret their performance.
- Solve thermodynamic problems systematically with correct use of units and property data.

Assessment and Grading

Component	Weighting	Description
Assignments and Quizzes	20%	Graded coursework issued and completed during the semester
Mid-Semester Test	10%	Written test covering the first half of the course content
Final Examination	70%	End-of-semester written examination

Note: this is a sample document generated for project testing purposes. Assessment weights and descriptive text are illustrative.